

Helcio Felipe

I am a physicist working at the interface of network science and information theory, developing principled methods to infer and summarize the structure of complex systems.

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Education

- Ph.D. Network Science.** Central European University, Austria 2022–2026
Thesis: *Summarizing network structures via information theory*
Supervisors: Federico Battiston and Alec Kirkley
- M.Sc. Physics.** Federal University of Rio Grande do Norte, Brazil 2019–2021
Supervisor: Gandhimohan Viswanathan
- B.Sc. Physics.** Federal University of Santa Catarina, Brazil 2012–2018

Publications

- H. Felipe**, A. Kirkley, and F. Battiston. “Summarizing higher-order network structures via information theory” in *Social Networks Beyond Pairwise Interactions* (Springer Computational Social Science Series, 2026).
-  **H. Felipe**, A. Kirkley, and F. Battiston. “Information theory for hypergraph similarity”. *Sci. Adv.* **12** (2026).
-  A. Kirkley, **H. Felipe**, F. Malizia, and F. Battiston. “Hypergraph backboning”. *arXiv:2606.00893* (2026).
- T. C. dos Santos, D. B. De Araujo, **H. Felipe**, J. G. V. Miranda, F. Palhano-Fontes, R. S. do Rosário, H. Onias, A. Viol, G. M. Viswanathan, and F. A. N. Santos, “An exploratory persistent-homology analysis of resting-state fMRI functional connectivity under Ayahuasca”. *Chaos Solit. Fractals* **209** (2026).
-  A. Kirkley, **H. Felipe**, and F. Battiston. “Structural Reducibility of Hypergraphs”. *Phys. Rev. Lett.* **135** (2025) [Editors’ Suggestion].
-  **H. Felipe**, F. Battiston, and A. Kirkley. “Network mutual information measures for graph similarity”. *Commun. Phys.* **7** (2024).
-  **H. Felipe**, A. Viol, D. B. De Araujo, M. G. E. Da Luz, F. Palhano-Fontes, H. Onias, E. P. Raposo, and G. M. Viswanathan. “Threshold-free estimation of entropy from a Pearson matrix”. *Europhys. Lett.* **141** (2023).

Research

- Doctoral Researcher.** Vienna, Austria 2022–2026
Network science and information theory, developing principled and efficient methods to compare and compress hypergraphs, and applying them to various empirical datasets using statistical inference and modeling.
- Graduate Research Assistant.** Natal, Brazil 2019–2021
Statistical physics and neuroscience of psychedelic-induced brain states, developing a threshold-free method of estimating the entropy of correlation matrices from neuroimaging time series.

Presentations

Invited talks

-  **TopoNets Satellite at NetSci 2026**, Boston, USA Jun 2026
Structural reducibility of hypergraphs
- CENTAI Seminar Series**, Turin, Italy Oct 2024
Network mutual information measures for graph similarity

Contributed talks

-  **NetSci 2026**, Boston, USA Jun 2026
Information theory for hypergraph similarity
-  **StatPhys 29 Intl. Conference on Statistical Physics**, Florence, Italy Jul 2025
Network mutual information measures for graph similarity
- Conference in Complex Systems (CCS) 2024**, Exeter, UK Sep 2024
Network mutual information measures for graph similarity

Teaching*

Introduction to Complex Systems (undergraduate)	2025
Advanced Topics in Applied Statistics (graduate)	2024
Computational Neuroscience (undergraduate)	2024
Classical Mechanics (undergraduate)	2020
Calculus & Physics (for the visually impaired, undergraduate)	2018

*I have a certificate of teaching from the Yehuda Elkana Center of CEU (2024).

Service and outreach

Journal referee for *Sci. Rep.* and *Chaos Solit. Fractals* (2026–present). Organization of the 10th Edition of “Data Stories” of the Department of Network and Data Science at CEU (Vienna, 2024). Technical support of NetSci 2023 (Vienna, 2023).

Fellowships and awards

- NetSci Educational Video (NetSci 2024)** 2024
Third place at the NetSci 2024 Educational Video competition
- High Impact Scholarship Award (University of Exeter)** 2021
Scholarship for the Exeter School on Urban Analytics
- Graduate scholarship (CAPES-Brazil)** 2019–2021
Master’s scholarship for research in statistical physics and complex systems.

References

Prof. Federico Battiston 
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Prof. Alec Kirkley 
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